BTS–III(S)–04.19 –0804 Reg. No. [ ]

C

B.Tech. Degree III Semester Supplementary Examination April 2019

CS 15-1306 DATA AND COMPUTER COMMUNICATION

(2015 Scheme)

Time : 3 Hours Maximum Marks : 60

PART A

(Answer ***ALL*** questions)

(10 × 2 = 20)

I. (a) Differentiate analog and digital signals with examples.

(b) The signal to noise ratio of a telephone line is 36 and the channel bandwidth is 2MHz. Find out the channel capacity?

(c) Explain the most common technique to change an analog signal to digital signal.

(d) Five channels, each with a 100kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

(e) Write down any four characteristics of virtual circuit networks.

(f) List the steps involved in creating checksum.

(g) How does a single bit error differ from a burst error?

(h) What are the four types of Frequency based coding schemes?

(i) Explain about piconet and scaternet?

(j) What are the categories of Fast Ethernet ?.

PART B

(4 × 10 = 40)

II. (a) Explain TCP/IP reference model in networking. (5)

(b) How do attenuation, distortion and noise impair a signal? (5)

OR

III. (a) Explain different line coding schemes. (6)

(b) What is signal to noise ratio? The power of a noise is 10 mw and the power of a noise is 1 μw. What are the values of SNR and SNR_{DB} (4)

IV. Explain about different transmission medium for communication. (10)

OR

V. (a) What is switching? Explain any three method for switching. (6)

(b) What is LATA? What are inter LATA and intra LATA services (4)

VI. Explain CRC with example. (10)

OR

VII. (a) With the help of diagram explain Go back N ARQ and Selective Repeat ARQ. (8)

(b) What is Hamming distance? What is the minimum Hamming distance? (2)

VIII. Explain various network topologies with diagrams. (10)

OR

IX. (a) Briefly explain about various connecting devices used in networking. (6)

(b) Briefly explain IEEE 802.11 (4)

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